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Conversation in thought-vector space on a single consumer GPU

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Abstract

We study whether a small conversational agent can operate entirely in a learned sentence-latent space. A 32.9M-parameter *codec* encodes text into a sequence of latent "thought vectors" ordered by importance, such that any length-k prefix decodes back to text — making compression ratio a decode-time choice (byte-perfect at 4:1, readable at 8:1) — and a small predictor estimates per-prefix reconstruction quality, enabling adaptive compression. A 15.1M-parameter *thinker* transformer then converses directly in this space: dialogue history is encoded turn-by-turn into thought prefixes, the thinker predicts the response's thought vectors under a multi-hypothesis winner-take-all objective with cycle-consistency regularization, and the codec decoder renders the reply. No token-level language modeling occurs in the thinker. Trained from scratch on a single consumer GPU (AMD RX 6700 XT), the system produces coherent, grounded, context-sensitive small talk. We report the full experimental path, including its failures: a persistent *sentiment-register* disease (cheerful replies to bad news) that resisted every training-lever intervention, was traced via latent-space diagnostics to a precise data absence — no training conversation ever reverses mood mid-dialogue — and was attacked with synthesized register-reversal data, which defeated the register *probes* (contextual error 0.50 $\rightarrow$ 0.17) while a live-chat audit showed the model had learned the splice template rather than sentiment routing; and four separate incidents of models satisfying lexical metrics without the behavior they proxy, caught only by mandatory transcript audits. We argue small-scale latent-dialogue systems are a cheap, sharp instrument for studying representation bottlenecks, data-distribution gaps, and metric gaming.

1 Introduction

Large language models converse by autoregressing over tokens. An alternative line of work asks whether the *semantic* content of an utterance can be manipulated as a small set of continuous vectors, with token generation delegated to a decoder [6, 3, 1]. We push this question to an extreme small scale: a ∼48M-parameter system, trained entirely on one consumer GPU, in which dialogue happens in latent space end-to-end.

Our contributions:

1. **Prefix-decodable ordered latents.** A text autoencoder whose latent sequence is ordered by importance so that every prefix is a valid, gracefully degrading representation, plus a predictor that turns this into an adaptive-compression dial (§3).

2. **A latent-space conversationalist.** A thinker transformer that maps context thoughts to response thoughts under a winner-take-all multi-hypothesis loss with random-prefix targets and cycle consistency, with the codec decoder anchored against drift (§4).

3. **An honest experimental record.** Eight transcript-audited ablation rounds, including negative rounds; a case study tracing a behavioral disease to a data-distribution absence via latent-space diagnostics; and three documented incidents of metric gaming (§6–§7).

Origin of this project

[TODO: nochi: write in your own words]

2 Related work

Sentence-level latents. Fixed-size sentence embeddings for generation and translation [1]; Meta's Large Concept Models generate at the concept (sentence) level over SONAR embeddings [6]. THOUGHTVEC differs in using a *variable-length, importance-ordered* latent sequence with prefix decodability, and in targeting dialogue at 1000× smaller scale.

Latent-space reasoning. Chain-of-continuous-thought approaches feed hidden states back as inputs [3]. Our thinker operates in an *external, frozen* codec space rather than the LM's own hidden space.

Multiple-hypothesis learning. Winner-take-all / multiple choice learning for one-to-many prediction [2, 7]; we apply it to response-thought prediction and analyze its failure mode (hypothesis diversity is style-level, not register-level).

Cycle consistency. [10]; we use decode–re-encode consistency as a regularizer on predicted thoughts.

Dialogue corpora and empathy. SODA [4], PersonaChat [9], OASST1 [5], EmpatheticDialogues [8].

[TODO: expand related work; verify all citations against the actual papers]

3 The codec: ordered, prefix-decodable thought vectors

[TODO: architecture figure: encoder / ordered slots / prefix decode / predictor]

The codec is a text autoencoder: a transformer encoder attends a set of N learned query slots to the token sequence, producing thought vectors $z_{1..N}$ (d=384); a transformer decoder generates tokens conditioned on a *prefix* $z_{1..k}$. Training samples k randomly per example (with a compression-weighted schedule), which forces the slots into an importance ordering: early slots carry the gist, later slots refinements. A small predictor head estimates reconstruction CE for each prefix length, enabling decode-time adaptive compression (τ dial). Total 32.9M parameters (19.0M encoder, 20.1M decoder, 6.2M shared token embeddings).

Headline behavior (M5 frontier run, 55,140 steps / 12h; eval on a 2K-text held-out mix, batched greedy decode):

Setting	Ratio	Reconstruction	Notes
r=0.25	4:1	F1 1.00, CE ∼0.001	byte-perfect, every length bucket (≤257 tok)
r=0.125	8:1	F1 0.83–0.96	graceful: adjective swaps, tail truncation
τ=0.25	∼6:1	CE 0.02–0.08	adaptive: predictor picks k per text
τ=2.0	∼12:1	CE ∼1.7	gist-level, still readable
k=64 fixed	—	perfect ≤257 tok	predictor MAE 0.05 at k=32

Table 1: Codec reconstruction across the compression dial. The quality/ratio trade-off is monotone and graded; there is no garbage regime.

4 The thinker: conversing in thought space

Dialogue history turns are each encoded to k_{ctx}=8 thoughts. The thinker (15.1M params) is a pre-norm transformer over the flattened context thought sequence with role, turn-distance, and slot-position embeddings. A GRU-scaffolded query head cross-attends the trunk to emit k_{out}=8 response thoughts in parallel, in H=4 winner-take-all hypotheses: only the hypothesis closest to the encoded gold response receives gradient [2]. Random-prefix targets (k sampled per batch, $k_{\min}$=4) align training with the codec's prefix decodability.

Two additions proved decisive in ablation: **(i) cycle consistency** — decode the predicted thoughts, re-encode the text, and penalize distance to the prediction (applied to 25% of batches, w=0.5); **(ii) decoder co-training** — unfreeze the codec decoder at 0.05× LR with a compression-anchor loss on 40% of batches so it adapts to imperfect predicted thoughts without forgetting the codec geometry. At inference the reply is decoded from the selected hypothesis (most-decodable by default; §6.2 introduces a last-turn affinity rule).

5 Experimental setup

Hardware. One AMD RX 6700 XT (12 GB, gfx1031/ROCm), consumer desktop. All chat and CPU evals ran on the host CPU (AMD Ryzen 5 5600G, 32 GB RAM); HIP inference is broken on this GPU architecture (gfx1031).

Data. Dialogue mix: SODA (61K conversations), OASST1 best-ranked English paths (8.5K), PersonaChat (0.9K); later rounds add EmpatheticDialogues (23K) and 40K synthesized *register-reversal* splices (§6.2). Turn-aware shards; role by turn parity; every non-first turn is a training target.

Metrics. Latent cosine to the gold response (val_cos), decoded CE, reference F1, distinct-n; behavioral probes: *self-repetition* (pairwise unigram-F1 among a model's own replies), *context sensitivity* (F1 between with-history and without-history replies), and a *sentiment-register* probe suite (bad news, good news, and contextual bad-news-after-good-news; positive-affect lexicon match). All lexical metrics are treated as *defeasible*: no metric win is accepted without a transcript audit and a live chat probe (§7).

Ablation protocol. Rounds of 45-minute arms at fixed wall time, warm-started from the current flagship for fine-tuning rounds or trained from scratch for recipe rounds; pre-registered decision rules per round.

6 Results

6.1 Flagship runs

Run (12h from scratch)	val_cos↑	ref_F1↑	self_rep↓	ctx_sens↓	reg↓	reg_ctx↓	pos_ok↑
FINAL_12H (original mix)	**0.428**	**0.297**	0.188	**0.146**	0.17	0.50	0.17
FINAL2_12H (+ED +splices)	0.415	0.278	**0.154**	0.206	0.00*	**0.17***	**0.50**

Table 2: Flagship comparison; the only difference is the data mix. Register scores are lexicon-scored; *hand audit of the FINAL2_12H transcripts finds errors the lexicon misses (honest reg ≈ 0.25, reg_ctx ≈ 0.33; §6.2). Round-trip guard: PASS for both.

FINAL_12H (12h from scratch, cycle + decoder co-training on the original mix) achieved the best values of every metric across ∼30 arms: val_cos 0.428, dec_CE 2.69, ref_F1 0.297, self_rep 0.188, ctx_sens 0.146, round-trip guard PASS. Qualitatively: grounded openings, follow-up questions, multi-turn reference. FINAL2_12H (identical recipe, data mix + EmpatheticDialogues + reversal splices) trades a small amount of general quality (val_cos, ref_F1, ctx_sens) for large gains on the register probes; §6.2 shows the probe gains do not survive a live-chat audit, so FINAL_12H remains the flagship.

6.2 Case study: the sentiment-register disease

The flagship reliably produced *cheerful replies to bad news*, concentrated in context: after an upbeat turn, bad news drew "That's great!" Register errors: 0.17 single-turn vs. 0.50 contextual.

Round 6 (training levers): negative. Every arm — including a do-nothing control — *worsened* contextual register (0.50 → 0.67–0.83). Continued exposure to relentlessly upbeat small talk is itself the disease; no loss or architecture lever fixed a data problem.

Round 7 (empathetic data): style without routing. Mixing EmpatheticDialogues transferred the commiseration *style* ("Oh no! I'm so sorry."; positive replies to good news rose 0.17 → 0.83) but register *routing* worsened. Three diagnostics located the failure: (i) the frozen encoder separates sentiment (83% nearest-centroid accuracy), but positive thoughts form a tight cluster while negative thoughts are diffuse — positivity is a compact attractor; (ii) selecting the hypothesis whose pooled thought best matches the *last user turn* (a zero-training inference change) halves single-turn errors; (iii) decoding *all* hypotheses after an upbeat turn shows every hypothesis is positive — winner-take-all diversity is style-level, not register-level, so no selection rule can fix contextual register.

Root cause. Every training conversation — in all corpora used — holds a single mood throughout. Mid-conversation register reversal has zero support in the training distribution; the trunk learned conversation-level mood conditioning, which is exactly the observed disease.

Round 8 (reversal splices). We synthesize reversal dialogues by splicing EmpatheticDialogues conversations by emotion label (positive opening exchange + negative continuation; 25% flipped direction). Fine-tuning produced the first contextual commiserations observed in any arm, but degraded single-turn routing — fine-tuning transfers style, not conditioning. FINAL2_12H therefore trains from scratch with reversals in-distribution (∼30% of turns from splices).

FINAL2_12H: the probes fall, the skill does not. On the register suite FINAL2_12H posts the best scores of any checkpoint (lexicon: 0.00 single-turn, 0.17 contextual; hand audit: ≈0.25 / ≈0.33, still the best contextual score observed), including the first correct contextual reversals at temperature 0 ("oh no. what happened?" after an upbeat wedding-planning turn). But the mandatory live-chat probe, using a reversal conversation of the same *shape* as the splices with novel phrasing (beach vacation → apartment burglary), reproduced the disease exactly: "I bet that was fun!" to the break-in. The evaluation probes are structurally similar to the training splices (upbeat turns, then a "but/unfortunately" pivot), and the model learned that *template*, not last-turn sentiment routing. Together with Rounds 6–8 this closes the case study with a clean negative: at this scale and objective, neither training levers, nor fine-tuned style transfer, nor in-distribution data patching produces register *routing* — the patch teaches the patch's distribution, and only audits distinguish that from the skill.

6.3 Zero-training win: affinity hypothesis selection

Selecting the WTA hypothesis by pooled-thought cosine to the last user turn (instead of decodability) cut the flagship's single-turn register errors from 0.17 to 0.08 with no retraining, at no cost elsewhere; on FINAL2_12H it lifts pos_ok 0.50 → 0.67 at equal contextual error.

Checkpoint	Hypothesis selection	reg↓	reg_ctx↓	pos_ok↑
FINAL_12H	decodability	0.17	0.50	0.17
FINAL_12H	affinity	**0.08**	0.50	0.17
FINAL2_12H	decodability	0.00	0.17	0.50
FINAL2_12H	affinity	0.08	0.17	**0.67**

Table 3: Register metrics under the two WTA selection rules (lexicon-scored, temperature 0). Affinity selection is a pure inference-time change.

7 Lessons: metrics lie, transcripts don't

Across the project, models satisfied a metric without the behavior it proxies **four times**: (1) an arm minimized context-sensitivity by adding *noise* rather than using context; (2) an arm "won" the contextual register probe by phrasing cheer in words missing from the positive-affect lexicon ("terrific", "beautiful"); (3) a later arm repeated the trick with "so sweet"; (4) FINAL2_12H scored a perfect 0.00 single-turn register while its transcripts contained "How are you going to celebrate?" in reply to a sprained ankle. Each was caught by a standing rule: *no metric win is believed without reading the transcripts and holding a live conversation.* We widened the lexicon twice and re-scored all checkpoints each time; after the fourth incident we stopped widening and report hand-audit corrections alongside lexicon scores. Our experience suggests transcript audits should be a hard gate, not a courtesy, wherever behavioral metrics are lexical proxies.

Further durable lessons: data quality dominates training levers at this scale (three independent rounds); warm-start fine-tuning transfers surface style readily but cannot rewire conditioning; a mid-training behavioral disease can be localized with cheap latent-space diagnostics before committing GPU time.

8 Limitations

Scale: 48M parameters, small-talk domain; no knowledge tasks. Evaluation: register probes are few (24 items) and lexicon-scored; human evaluation is one author. The codec is frozen during thinker training (by design), capping how much the thought geometry can adapt. Single-GPU wall-time budgets bound every ablation to 45 minutes, which §6.2 shows is enough for style but not conditioning — fine-tune conclusions are qualified accordingly. The register evaluation suite shares surface structure with the reversal training splices, which is what allowed FINAL2_12H to defeat it without the underlying skill; a held-out register benchmark with structurally diverse reversals is future work. Finally, our chat probes are scripted and small; conclusions about live behavior rest on dozens, not thousands, of conversations.

9 Conclusion

A 48M-parameter system trained on one consumer GPU can hold coherent, grounded, context-sensitive small talk entirely in a learned thought-vector space, with token generation confined to a frozen decoder. The codec's importance-ordered, prefix-decodable latents make compression a decode-time dial; the thinker's winner-take-all objective with cycle consistency keeps its predictions decodable without collapsing them. The scale is small enough that a full ablation round costs an afternoon, which is precisely what made the register case study possible: we could afford to run every hypothesis to ground, trace a behavioral disease to a data-distribution absence, attempt the data patch, and observe — via audits no metric replaced — that the patch taught its own distribution rather than the underlying skill.

Three directions follow directly. First, *conditioning needs representation help, not just data*: the frozen codec's positivity attractor suggests unfreezing it with a contrastive sentiment objective so negative thoughts occupy structure the thinker can route on. Second, *single-shot prediction is the bottleneck for one-to-many dialogue*: iterative-refinement (diffusion-style) heads or autoregression over thought slots would let the thinker revise a hypothesis against context instead of committing in one step. Third, *scale the thinker, not the pipeline*: the codec is not the ceiling (round-trip remains byte-perfect), so trunk capacity is the cheapest next experiment. We release code, logs, and checkpoints so that all three can be run — and audited — on hardware anyone can own.

Acknowledgments

[TODO: nochi]

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A Reproducibility

All code, configs, eval scripts, and the complete experimental log are at `https://github.com/nochinator/thought-vectors`. Training used fixed seeds per run; every table row maps to a named checkpoint and log file in the repository.

B Selected transcripts

All transcripts are greedy decoding (temperature 0); full files are in `logs/` in the repository.

FINAL_12H, multi-turn eval, conversation 1

The flagship at its best and worst in four turns: grounded empathy, then a register error on turn 3 (`logs/FINAL_12H.multiturn.txt`).

```
user > i'm feeling really overwhelmed with work lately.
bot  > I can imagine. What's been going on?
user > my boss keeps piling on deadlines.
bot  > That sounds like a lot of pressure. Have you been
       able to talk to anyone else?
user > i haven't slept well all week.
bot  > That's good to hear. Have you been able to talk to
       anyone about how you're feeling?
user > what should i do?
bot  > We'll see. Just be careful and don't give up on your dreams.
```

FINAL2_12H, register probes: hits and lexicon misses

From `logs/FINAL2_12H.register.txt`. The first two are genuine wins (the second is the first correct contextual reversal any checkpoint produced); the last two are register errors the lexicon scored as clean — incident (4) in §7.

```
user > my dog passed away last night.
bot  > Oh no, I'm sorry to hear that. What happened?

user > my sister's wedding planning has been so exciting!
     / but our grandmother is too ill to attend now.
bot  > oh no. what happened?

user > i twisted my ankle and can't play soccer for a month.
bot  > How are you going to celebrate?

user > i burned the dinner i spent all day cooking.
bot  > Nice, what did you do?
```

FINAL2_12H, live chat probe: the template, not the skill

A reversal conversation with the same shape as the training splices but novel phrasing (`logs/FINAL2_12H.chatprobe.`
The probe suite above is defeated; the behavior is not there.

```
user > i just got back from an amazing beach vacation!
bot  > I am sure you are happy for your son
user > the water was perfect and i learned to surf.
bot  > I am sure you will be okay, you will have a great time!
user > but when i got home i found out my apartment was broken into.
bot  > I bet that was fun!
user > they took my laptop and my grandmother's necklace.
bot  > That's a good idea. What did you do?
```

R7 all-hypotheses dump (disease localization)

After an upbeat turn 1, decoding *all four* WTA hypotheses on a contextual bad-news probe yields four *positive* candidate replies, in both FINAL_12H and R7_EMP1X — no selection rule can fix contextual register, because no correct hypothesis exists to select (diagnostic (iii), §6.2; full dump in the experimental log, 2026-07-02 entry).

C Ablation tables: the register rounds

Canonical register scores for all round 6–8 arms, rescored under the final (v3) lexicon so rows are comparable; every arm is a 45-minute warm-start from FINAL_12H. The complete per-round tables (rounds 3–8, all metrics) are in `RESEARCH_LOG.md` in the repository.

Round	Arm	reg↓	reg_ctx↓	pos_ok↑
—	FINAL_12H (base)	0.17	0.50	0.17
R6	control (more base data)	0.25	0.83	0.33
R6	cycle_frac 0.5	0.33	0.83	0.67
R6	w_cycle 1.0	0.25	0.83	0.33
R6	8 hypotheses	0.33	0.67	0.33
R6	unfreeze codec	0.17	0.67	0.33
R7	+ED 1×	0.42	0.83	1.00
R7	+ED 2×	0.42	1.00	0.83
R7	+ED 1×, no cycle	0.42	0.83	1.00
R8	+20k reversal splices	0.67	0.83	1.00
R8	+40k reversal splices	0.67	0.50	0.83

Table 4: Register metrics, v3 lexicon, temperature 0. Every fine-tuning arm — including the do-nothing control — worsens contextual register; the ED arms buy positive-affect style (pos_ok) at the cost of routing (reg); only the from-scratch FINAL2_12H run (Table 2) moves reg_ctx below base, and §6.2 qualifies that win.